COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
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ACTIVITY NO. 4.1 STACKS

1. PROCEDURE OUTPUT

ILOB2

OUTPUT

```cpp
#include <iostream>
#include "stackList.h"

int main()
{

    std::cout << "--- PUSHING 'c' 'p' 'e' ONTO THE STACK ---\n";
    // push
    push('c');
    push('p');
    push('e');

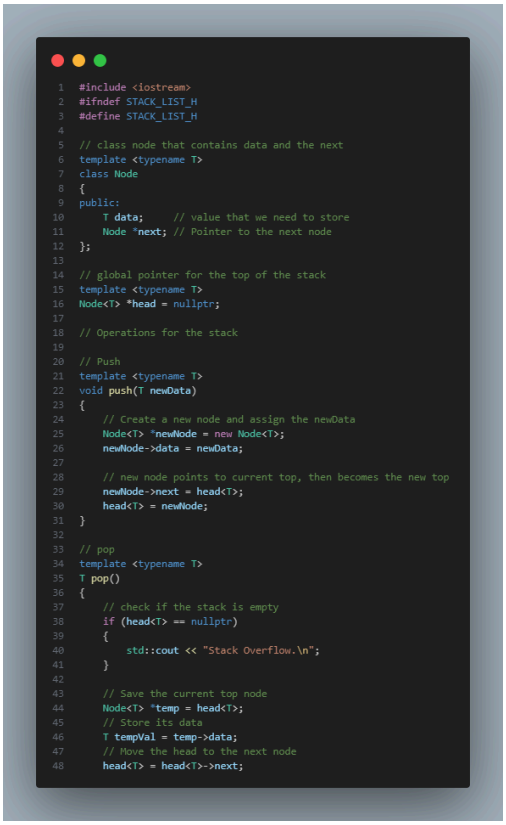
    // Display the current tag
    Top<char>(); // Top should be 'e'

    displayAll<char>();
    std::cout << "\n --- Popping all the elements ---\n";

    while (isEmpty<char>())
    {
        std::cout << "Popped: " << pop<char>() << std::endl;
    }

    std::cout << "\n --- Testing the Underflow ---\n";
    pop<char>();

    return 0;
}
```



```cpp
#include <iostream>
#ifndef STACK_LIST_H
#define STACK_LIST_H

// class node that contains data and the next
template <typename T>
class Node
{
public:
    T data;     // value that we need to store
    Node *next; // Pointer to the next node
};

// global pointer for the top of the stack
template <typename T>
Node<T> *head = nullptr;

// Operations for the stack

// Push
template <typename T>
void push(T newData)
{
    // Create a new node and assign the newData
    Node<T> *newNode = new Node<T>;
    newNode->data = newData;

    // new node points to current top, then becomes the new top
    newNode->next = head<T>;
    head<T> = newNode;
}

// pop
template <typename T>
T pop()
{
    // check if the stack is empty
    if (head<T> == nullptr)
    {
        std::cout << "Stack Overflow.\n";
    }

    // Save the current top node
    Node<T> *temp = head<T>;
    // Store its data
    T tempVal = temp->data;
    // Move the head to the next node
    head<T> = head<T>->next;
```



```cpp
    delete temp;
    // Returnrn the removed value
    return tempVal;
}

// isEmpty
template <typename T>
bool isEmpty()
{
    return head<T> == nullptr;
}

// Top
template <typename T>
void Top()
{
    // Error catching
    if (isEmpty<T>())
    {
        std::cout << "Stack Underflow.\n";
        return;
    }

    // Display the Top node
    std::cout << "Top of the Stack: " << head<T>->data << "\n";
}

// Display All
template <typename T>
void displayAll()
{
    if (head<T> == nullptr)
    {
        std::cout << "Stack Underflow.\n";
        return;
    }

    std::cout << "Displaying the Stack: \n";
    Node<T> *temp = head<T>;
    while (temp != nullptr)
    {
        std::cout << temp->data << std::endl;
        temp = temp->next;
    }
    std::cout << std::endl;
}

#endif
```

```
--- PUSHING 'c' 'p' 'e' ONTO THE STACK ---
Top of the Stack: e
Displaying the Stack:
e
p
c


 --- Popping all the elements ---

 --- Testing the Underflow ---
```

FUNCTION	DESCRIPTION
push(int newData)	Creates a new node and stores the newData init then inserts it at the top of the stack.
pop()	Removes the node at the top of the stack and returns its data value. If the stack = null it prints an error code (Stack Underflow) then returns -1. Otherwise it deletes the nodes one by one starting from the head node which is the top of the stack going down to the bottom of the stack.
top()	Prints the value stored in the node at the top of the stack. If there is no value it prints an error

	(Stack is empty)
displayAll()	Displays the value of each node until the next temp node equals to null, if the temp node equals to null it stops the loop. If the stack is empty it prints “Stack Underflow.”

4. SUPPLEMENTARY ACTIVITY

ILO C: Solve problems using an implementation of stack:

The following problem definition and algorithm is provided for checking the balancing of symbols. Create an implementation using stacks. Your output must include the following:

a. Stack using Arrays
b. Stack using Linked Lists
c. (Optional) Stack using C++ STL

Problem Definition:
Stacks can be used to check whether the given expression has balanced symbols. This algorithm is very useful in compilers. Each time the parser reads one character at a time. If the character is an opening delimiter such as (, {, or [- then it is written to the stack. When a closing delimiter is encountered like), }, or]-the stack is popped. The opening and closing delimiters are then compared. If they match, the parsing of the string continues. If they do not match, the parser indicates that there is an error on the line.

Steps:
1. Create a Stack.
2. While(end of input is not reached) {
    a) If the character read is not a symbol to be balanced, ignore it.
    b) If the character is an opening symbol, push it onto the stack.
    c) If it is a closing symbol:
        i) Report an error if the stack is empty.
        ii) Otherwise, pop the stack.
    d) If the symbol popped is not the corresponding opening symbol, report an error.
3. At the end of input, if the stack is not empty: report an error.

Self-Checking:
For the following cases, complete the table using the code you created.

Expression	Valid? (Y/N)	Output (Console Screenshot)	Analysis
(A+B)+(C-D)			
((A+B)+(C-D)			
((A+B)+[C-D])			

((A+B]+[C-D]}			

Tools Analysis:

- How do the different internal representations affect the implementation and usage of the stack?

5. Conclusions/Learnings:

Provide the following:

- Summary of lessons learned
- Analysis of the procedure
- Analysis of the supplementary activity
- Concluding statement / Feedback: How well did you think you did in this activity? What are your areas for improvement?

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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